# Introduction, Exponential Smoothing & Forecast Evaluation

## ECON 8310: Business Forecasting — Lecture 1

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# Lecture Outline

# Why Forecasting Matters

Every business decision made today depends on expectations about tomorrow.

# Forecasting Is Everywhere in Business

| Domain | The question someone is actually asking |
|---|---|
| Operations & supply chain | How much inventory should Walmart order for Black Friday? How many nurses does a hospital need next Tuesday night? |
| Finance & risk | Will credit card defaults rise next quarter? What will the Fed Funds rate be in six months? |
| Marketing & strategy | How many units will this new product sell in year one? What is the lifetime value of this customer cohort? |
| Macro & public policy | Will GDP grow by 2% or 3% next year? What will unemployment be in Q4? |

Every one of these is a *decision* waiting on a number. **A decision is only as good as the forecast it rests on.**

# The Basic Setup

> **Time Series**
>
> A **time series** is an ordered sequence of observations $\{y_1, y_2, \ldots, y_T\}$ where $t$ indexes time.

**Notation used throughout the course:**

- $y_t$ — observed value at time $t$
- $h$ — **forecast horizon**, how many periods ahead
- $\hat{y}_{t+h|t}$ — forecast of $y_{t+h}$ made using information at time $t$
- $e_t = y_t - \hat{y}_{t|t-1}$ — **forecast error**

Forecasting monthly retail sales, $h = 1$ is next month and $h = 12$ is a year out. The problem gets harder as $h$ grows, because uncertainty compounds. Under squared-error loss the optimal forecast is the **conditional expectation**, $\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} \mid \mathcal{F}_t]$, where $\mathcal{F}_t$ is everything known at time $t$.

# Components of a Time Series

Any series can be decomposed into four parts (additive form): $y_t = T_t + S_t + C_t + I_t$.

- $T_t$ **Trend** — long-run direction
- $S_t$ **Seasonality** — regular, calendar-driven pattern
- $C_t$ **Cycle** — medium-run business-cycle swings
- $I_t$ **Irregular** — random shocks

Which parts matter depends entirely on the series: retail sales are trend + seasonality, quarterly GDP is trend + cycle, daily stock returns are mostly irregular, and energy demand has all four.

**Ignoring seasonality causes large, systematic errors.** A non-seasonal model on monthly retail data will consistently under-forecast December and over-forecast January — every single year.

# Always Beat a Benchmark First

Before presenting any forecast model, check that it beats a **naive benchmark**. A model that cannot beat the baseline has zero value.

| Benchmark | Forecast | Use when |
|---|---|---|
| Naïve | $\hat{y}_{t+h\|t} = y_t$ | Asset prices, random walks |
| Seasonal naïve | $\hat{y}_{t+h\|t} = y_{t+h-m}$ | Any series with a season |
| Mean | $\hat{y}_{t+h\|t} = \bar{y}_t$ | No trend, no season |
| Drift | $\hat{y}_{t+h\|t} = y_t + h\hat{c}$ | Trend, no season |

This is not a formality. In the M4 Competition many machine learning models *failed* to beat simple exponential smoothing (Makridakis et al. 2020).

# Exponential Smoothing

Recent observations should carry more weight than old ones.

# The Problem with Equal Weights

Most classical methods treat every historical observation equally — but what if the world **changes**? When a retailer's sales pattern shifts after a store renovation, equal-weighting the pre-renovation data buys months of systematic over- or under-forecasting. **Exponential weighting** assigns geometrically declining weights, so recent observations matter more:

| Observation | Age | Weight |
|---|---|---|
| $y_T$ | 0 | $\alpha$ |
| $y_{T-1}$ | 1 | $\alpha(1-\alpha)$ |
| $y_{T-2}$ | 2 | $\alpha(1-\alpha)^2$ |

$\alpha \in (0,1]$ is the forgetting rate. High $\alpha$ = short memory; low $\alpha$ = long memory.

# Simple Exponential Smoothing (SES)

### SES Update Equation

Let $\ell_t$ be the **level** at time $t$:

$$\ell_t \;=\; \alpha\, y_t \;+\; (1-\alpha)\,\ell_{t-1}, \qquad 0 < \alpha \leq 1$$

**Forecast:** $\hat{y}_{T+h|T} = \ell_T$ for all horizons $h \geq 1$ (flat)

**Large $\alpha$ (e.g. 0.8):**

- Reacts quickly to shocks
- Noisy, volatile forecasts
- Good for rapidly changing series

**Small $\alpha$ (e.g. 0.1):**

- Slow to adapt
- Smooth, stable forecasts
- Good for slowly evolving series

SES **nests both benchmarks**: $\alpha = 1$ gives the naïve forecast, and $\alpha \to 0$ gives the historical mean. *Use when there is no trend and no seasonality.*

# Holt's Method: Adding Trend

SES produces a *flat* forecast. If the series trends, track both the level and the slope.

**Holt's Linear Exponential Smoothing (Holt 2004)**

**Level:** $\ell_t = \alpha\, y_t + (1-\alpha)\,(\ell_{t-1} + b_{t-1})$

**Trend:** $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\, b_{t-1}$

**Forecast:** $\hat{y}_{T+h|T} = \ell_T + h\, b_T$

Read $\ell_t$ as where the series *is* now and $b_t$ as its current growth rate. **Linear extrapolation gets unrealistic at long horizons.** For $h > 6$ use the **damped trend** variant, $\hat{y}_{T+h|T} = \ell_T + \sum_{j=1}^{h} \phi^j b_T$ with $0 < \phi < 1$. At $\phi = 0.85$ the forecast converges to a finite ceiling instead of growing forever.

# Holt-Winters: Level, Trend, and Seasonality

### Holt-Winters Additive Method (Winters 1960)

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\, b_{t-1}$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)\, s_{t-m}$$

**Forecast:** $\hat{y}_{T+h|T} = \ell_T + h\, b_T + s_{T+h-m}$ ($m =$ seasonal period)

**Additive seasonality:**

- Seasonal swings are constant in size
- Same $\pm$ amount every December

**Multiplicative seasonality:**

- Swings grow with the level
- Retail sales, tourism

In US retail sales the December spike grows every year as overall sales grow, which makes it **multiplicative**. Requires at least $2m$ observations to initialize ($m = 12$ for monthly data).

# The ETS Framework: A Unified View

Hyndman and Athanasopoulos (2021) unify every exponential smoothing method as a **state space model**: Error × Trend × Seasonal.

| Component | Options | Meaning |
|---|---|---|
| Error ($E$) | A, M | Additive / Multiplicative |
| Trend ($T$) | N, A, $\mathrm{A_d}$ | None / Additive / Damped |
| Seasonal ($S$) | N, A, M | None / Additive / Multiplicative |

That gives 15 valid combinations, and the ones we have just built are among them: **ETS(A,N,N)** is SES, **ETS(A,A,N)** is Holt linear, **ETS(A,$\mathbf{A_d}$,N)** is Holt damped, **ETS(A,A,A)** is Holt-Winters additive, and **ETS(A,A,M)** is Holt-Winters multiplicative. **Automatic selection:** fit all 15 and pick the minimum AIC. In Python,

`statsmodels.tsa.exponential_smoothing.ets.ETSModel`.

# Forecast Evaluation

Forecasting without evaluation is not science — it is storytelling.

# How Do We Measure Forecast Accuracy?

Given out-of-sample errors $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$:

**Scale-dependent:**

$$\text{RMSE} = \sqrt{\tfrac{1}{H}\textstyle\sum_{h=1}^{H} e_{T+h}^2}$$

$$\text{MAE} = \tfrac{1}{H}\textstyle\sum_{h=1}^{H} |e_{T+h}|$$

Same units as $y_t$. Compare models on the *same* series.

**Scale-free:**

$$\text{MAPE} = \tfrac{100\%}{H}\textstyle\sum_{h=1}^{H} \left|\frac{e_{T+h}}{y_{T+h}}\right|$$

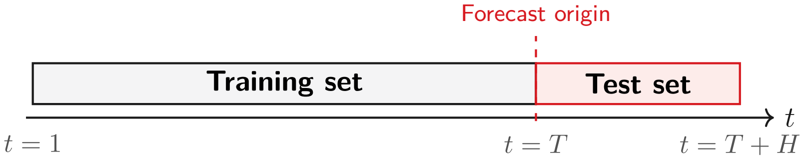
$$\text{MASE} = \frac{\text{MAE}}{\text{MAE}_{\text{seas. naïve}}}$$

Compare *across* series.

**RMSE vs. MAE:** RMSE punishes large errors harder. When big misses are especially costly — a stockout, an understaffed shift — prefer RMSE. **MASE** is the one to report across series: MASE $< 1$ beats seasonal naïve, MASE $> 1$ is worse than doing nothing clever. It is the standard in the M-Competitions (Makridakis et al. 2020).

# The Golden Rule: Always Evaluate Out-of-Sample

**Never evaluate a forecast model on the data used to fit it.** In-sample fit measures how well the model memorized the past, not how well it predicts the future.




**Never randomly shuffle time series observations.** Use `TimeSeriesSplit` in scikit-learn, never `KFold`. Shuffling lets future data train the model — data leakage that makes results look excellent and forecast badly.

# Course Roadmap

What we will cover, and how it fits together.

# ECON 8310: Four-Part Structure

| Part | Lectures |
|---|---|
| I. Time series | 1. Intro, ETS & evaluation ← today 2. ARIMA & VAR 3. Generalized additive models |
| II. Tree-based | 4. Decision trees 5. Tree ensembles: random forests & boosted trees 6. Regularization & model selection |
| III. Deep learning | 7. Neural networks (PyTorch) 8. CNN architectures 9. RNNs, LSTMs & Transformers |
| IV. Bayesian | 10. Foundations 11. Time series & hierarchical models 12. Bayesian linear regression |

Python throughout: statsmodels, scikit-learn, xgboost, PyTorch, prophet, pyGAM, pymc.

## Tools and What You Will Be Able to Do

**The Python stack:** `statsmodels` (ETS, ARIMA, VAR) · `scikit-learn` (models, pipelines, cross-validation) · `xgboost` (gradient boosting) · `torch` (neural networks) · `prophet` and `pygam` (GAMs) · `pymc` (Bayesian models). **By the end of the semester you will be able to:**

- ✓ Choose the right method for a given business problem
- ✓ Implement and tune it in Python
- ✓ Evaluate it rigorously out-of-sample
- ✓ Interpret the result for a non-technical stakeholder
- ✓ Use AI tools effectively to accelerate the analysis
- ✓ Avoid the most common forecasting mistakes

# Lecture 1: Key Takeaways

1. Forecasting underpins every forward-looking business decision. The optimal point forecast (under squared loss) is the **conditional expectation**.
2. **Always beat a benchmark first.** Naïve, seasonal naïve, mean, drift.
3. Exponential smoothing assigns **geometrically declining weights** to past observations. $\alpha$ controls how fast old data is forgotten.
4. SES (level only) $\rightarrow$ Holt (+ trend) $\rightarrow$ Holt-Winters (+ season). The **ETS framework** unifies all variants; AIC selects automatically.
5. **Evaluate out-of-sample only.** RMSE penalizes large errors; MASE benchmarks against seasonal naïve. Never shuffle time series data.

*Next: ARIMA & VAR — a regression-based approach to time series with rich autocorrelation structure.*

# References I


- Holt, Charles C. (2004). "Forecasting Seasonals and Trends by Exponentially Weighted Moving Averages". In: *International Journal of Forecasting* 20.1. Reprint of the 1957 ONR Research Memorandum, pp. 5–10.
- Hyndman, Rob J. and George Athanasopoulos (2021). *Forecasting: Principles and Practice*. 3rd. Open access. Melbourne, Australia: OTexts. URL: https://otexts.com/fpp3/.
- Makridakis, Spyros, Evangelos Spiliotis, and Vassilios Assimakopoulos (2020). "The M4 Competition: 100,000 Time Series and 61 Forecasting Methods". In: *International Journal of Forecasting* 36.1, pp. 54–74.
- Winters, Peter R. (1960). "Forecasting Sales by Exponentially Weighted Moving Averages". In: *Management Science* 6.3, pp. 324–342.